

BanglaCyberBench: A Dual-Script Benchmark and a Script-Aware Ensemble for Bengali Cyberbullying Detection

Sefayet Alam^{1,*}, Naim Parvez¹, A. F. M. Minhazur Rahman¹

¹Department of Computer Science and Engineering, Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh

*Corresponding author: Sefayet Alam; sefayetalam14@gmail.com

Abstract. Bengali harmful-language classifiers are usually evaluated on a random split of one source written in one script, which gives little evidence about behaviour on data collected elsewhere. We present BanglaCyberBench, a four-source benchmark for five-class harmful-language classification in Bangla and Romanized Bangla, and we evaluate a script-aware ensemble trained with cross-entropy and the Fast Gradient Method (FGM). On the in-domain test split the ensemble reaches a Macro-F1 of 0.8225, with a 95% bootstrap confidence interval of [0.8146, 0.8300]. Macro-F1 is 0.8303 for Bangla-script comments and 0.3900 for Romanized comments under the fixed five-class definition. The supported-class Macro-F1 for Romanized comments is 0.4875, because the Romanized test subset contains no *threat* examples. Source-held-out Macro-F1 ranges from 0.4612 to 0.5850, so performance falls when the test source is unseen during training. The labels type the harmful content of an isolated comment rather than cyberbullying in its interactional sense of repetition, power imbalance, or victim history, none of which the comment-level sources record.

Keywords: Bengali harmful-language detection; Romanized Bangla; dual-script benchmark; script-aware ensemble; adversarial fine-tuning; class imbalance; cross-source generalisation.

1. Introduction

Bengali harmful-language detection has to account for two written forms. Bengali is spoken by more than 240 million people, and a large share of its social-media text is typed in the Latin alphabet rather than in Bangla script. The Romanized form *tui akta faltu manush* and its Bangla-script equivalent carry the same insult in two different surfaces. Existing Bengali datasets differ in source, script, and label definition, which makes direct comparison difficult and complicates transfer between them.

Most prior studies report random within-dataset evaluation. Such evaluation does not measure performance on a new source, and cross-dataset degradation is already well established in abusive-language detection [1–5]. The published comparator for this study reports a Weighted-F1 of 0.8923 and a Macro-F1 of 0.8803 on its single-source protocol [6]. We therefore treat source-held-out evaluation as a Bengali replication and quantification of a known problem rather than as the discovery of source shift.

We address three research questions. **RQ1** (Section 3) evaluates whether four Bengali sources can be mapped to a shared five-class taxonomy. **RQ2** (Sections 6.1, 6.3 and 6.5) measures in-domain and script-stratified performance of a script-aware ensemble. **RQ3** (Sections 6.4 and 6.6) evaluates component sensitivity and source-held-out generalisation under the retained protocol.

1.1 Contributions

(C1) Shared benchmark protocol. We map four Bengali harmful-language datasets to a common five-class taxonomy, preserve script information, and construct random and source-held-out evaluation protocols. The merged corpus contains 94,323 comments after exact-text deduplication, 60.4% in Bangla script and 39.6% Romanized.

(C2) Script-aware reference model. We evaluate an ensemble of Bangla, bilingual, and multilingual encoders trained with cross-entropy and FGM. BanglaBERT is excluded from Romanized inputs according to its pretraining scope.

(C3) Multi-level evaluation. We report in-domain, per-class, script-stratified, component, taxonomy,

and source-held-out results. Bootstrap confidence intervals are reported where the required predictions are available.

(C4) Evaluation findings. The experiments quantify cross-source degradation and report script-stratified performance together with the support of each class.

The proposed model is offered as the fully specified reference system that the protocol needs, not as an algorithmic novelty claim. Pretrained encoders, FGM, and weighted fusion are all established components. The new material is the consolidated resource, the evaluation protocol applied to it, and the measurements that protocol produces.

The rest of the paper is organised as follows. Section 2 reviews prior work. Section 3 describes the benchmark and its construction. Section 4 formalises the proposed model on a single comment. Section 5 gives the experimental setup. Section 6 reports results. Section 7 discusses the findings and their limitations, and Section 8 concludes.

2. Related Work

2.1 Bengali Cyberbullying and Hate-Speech Detection

Earlier Bengali harmful-language studies evaluated classical classifiers, neural sequence models, and transformer encoders, usually on one dataset [7–10]. TF-IDF features with logistic regression, support vector machines, and tree ensembles were common in the earliest work [7, 10, 11], and BiLSTM and CNN-BiLSTM models with FastText or GloVe embeddings followed [12, 13]. Transformer encoders are now the usual choice. Emon et al. [14] and Aurpa et al. [15] applied BERT-family encoders to Bangla abusive-comment detection, Karim et al. [9] added explainability to Bengali hate-speech classification, and Saifullah et al. [16] combined deep learning with transformer language models. Recent transformer-based systems improve in-domain performance, but most do not evaluate transfer to an unseen source. The comparator used in this study evaluates a transformer ensemble on Facebook-44K under a single-source protocol [6]. That protocol does not measure cross-source transfer and does not use FGM, so our FGM result is a property of the proposed training configuration rather than a claim about the comparator.

Bengali resources have grown alongside the methods. Romim et al. [8] released a Bengali hate-speech corpus with baseline evaluation, and Ahmed et al. [17] built a Facebook harassment dataset with exploratory analysis. Regional and multi-task resources such as BIDWESH [18] and LLM-based multi-task hate-speech corpora [19] extend coverage to dialects and severity. BanTH [20], a large transliterated-Bangla dataset and one of the four sources consolidated here, and BanCyB [21] broaden the field further. Our claim is therefore scoped to this consolidated benchmark and its evaluation protocol rather than to Bengali harmful-language detection in general. Table 1 places BanglaCyberBench against representative prior resources on the axes that matter for this paper: number of sources, script coverage, taxonomy granularity, and whether transfer to an unseen distribution was measured.

Table 1. Representative Bengali abuse-detection resources against BanglaCyberBench. “Shift tested” asks whether the study measured transfer to an unseen source or script rather than only a random in-domain split.

Resource	Sources	Script	Classes	Shift tested	Size
Romim et al. [8]	1	Bangla	binary	no	30k
Ahmed et al. [7]	1	Bangla + Rom.	binary	no	12k
Karim et al. [9]	1	Bangla	4	no	8,087
Base paper [6]	1	Bangla	5	no	44,001
BanglaCyberBench (ours)	4	Bangla + Rom.	5	yes	94,323

2.2 Cross-Dataset Generalisation

Cross-dataset degradation is a recurring result in abusive-language detection. Prior work attributes it to dataset-specific sampling, annotation policy, topic distribution, and differences in the label space [1–4]. Yin and Zubiaga [5] review these obstacles and the mitigation strategies proposed for them. Our contribution is a Bengali, dual-script quantification under a fixed source-held-out protocol, not the first observation that random-split performance transfers poorly.

2.3 Romanized Bangla and the Script Problem

Romanized Bangla is non-standardised and frequently code-mixed with English, and it drops vowels freely. Ahmed et al. [7] is one of the few Bengali studies to treat both scripts side by side, and it does so with classical models on a modest scale. TB-OLID provides a directly relevant offensive-language benchmark for transliterated and code-mixed Bangla [22], BanglaTLit provides back-transliteration pairs [23], and BanTH provides a Romanized Bengali harmful-language resource [20]. The available pretrained encoders divide along the same line. BanglaBERT [24] is an ELECTRA-style discriminator pretrained primarily on Bangla-script text. BanglishBERT is a Bangla-English bilingual encoder and is not pretrained on Romanized Bengali. MuRIL [25] includes transliterated Indic text in pretraining, and XLM-R [26] provides a multilingual reference. Recent Bangla shared-task work also provides context for current hate-speech evaluation [27]. Whether one encoder can serve both scripts is an open question, and the script-aware design of Section 4 treats it as a pretraining-scope constraint rather than a settled empirical result.

2.4 Adversarial and Imbalance-Aware Fine-Tuning

Perturbing continuous word-embedding parameters during training is an inexpensive regulariser for small-data text classification. The Fast Gradient Method (FGM) [28], a text adaptation of the image-domain idea of Goodfellow et al. [29], adds one adversarial forward-backward pass per batch. Other components target class imbalance and overfitting. Focal loss [30] and class-balanced weighting [31] down-weight easy majority examples. Multi-sample dropout [32] reduces head variance, R-Drop [33] enforces consistency between two dropout passes, weight averaging in the style of mean teachers [34] stabilises evaluation weights, and logit adjustment [35] corrects the long tail at inference. Our fixed single-backbone sensitivity run compares these choices descriptively and does not establish a universal ordering. The proposed model uses cross-entropy with FGM as a transparent compromise between the measured differences and training cost.

3. The BanglaCyberBench Benchmark

BanglaCyberBench merges four public Bengali datasets into one deduplicated, dual-script, five-class corpus. Fig. 1 shows the construction pipeline, and this section describes each stage.

Task scope. BanglaCyberBench evaluates comment-level harmful-language classification. Each isolated comment is assigned to None, Abusive, Religious, Sexual, or Threat. The data do not contain conversation history, repeated targeting, power imbalance, or victim information. The task therefore does not model cyberbullying as an interactional phenomenon, and the term is retained only for continuity with the source datasets. A title of the form “harmonising multi-source Bengali harmful-language data” would describe the contribution equally well.

3.1 Sources and Schema Unification

The benchmark combines Facebook-44K, BanTH, Multilabel-12.5K, and the publicly available BD-SHS test partition (Table 2). The BD-SHS contribution contains 5,029 comments. We map the source texts, labels, provenance, and script information to a common representation before deduplication and splitting. BanTH supplies the Romanized comments, and the other three sources use Bangla script. BanTH is therefore the only Romanized source, which is why source-held-out evaluation on BanTH and any cross-script analysis are closely related rather than independent.

3.2 Preprocessing

Bengali harmful language relies on cues that aggressive normalisation destroys, so the cleaning is light and script-safe. We apply Unicode NFKC normalisation, remove invisible and zero-width characters, mask URLs and user mentions, preserve hashtag text, convert emoji to text tokens, collapse runs of three or more identical characters down to two, and standardise whitespace. Collapsing repeats keeps *baaaaje* and *baje* from being treated as unrelated. We do not case-fold beyond Unicode rules, strip punctuation, or remove stopwords, because emphatic repetition, punctuation, and emoji all carry signal and subword tokenisers handle the raw form well. The same procedure is applied before deduplication and before model input construction. Fig. 2 shows the composition of the resulting benchmark by

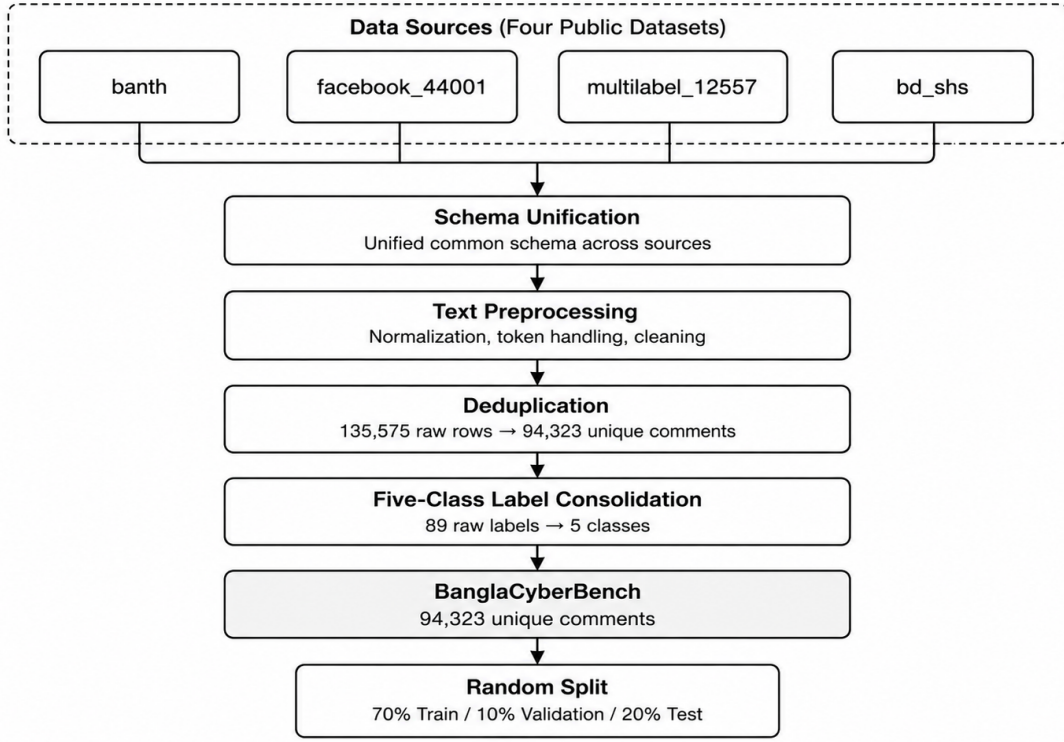


Figure 1. Benchmark construction pipeline. Four sources are mapped to a common representation, normalised, deduplicated on cleaned text (135,575 $\rightarrow$ 94,323), consolidated from 89 raw label strings into five classes by a priority rule, and split into a stratified random 70/10/20 partition. The source-held-out configurations are described in Section 3.5.

Table 2. The four sources after deduplication. BanTH is the only Romanized-Bangla source, and the other three are Bangla script. Shares are of the 94,323-comment benchmark.

Source	Script	Origin	Samples	Raw label style
Facebook-44K [17]	Bangla	Mendeley Data	43,078	single label column
BanTH [20]	Romanized	public release	37,334	binary column and per-type indicators
Multilabel-12.5K [36]	Bangla	Mendeley Data	8,882	separate per-type indicators
BD-SHS (public test partition) [37]	Bangla	ACL repository	5,029	one harmful column and one type column
Total	NA	NA	94,323	NA

source, script, and class.

3.3 Deduplication

Records are grouped by exact normalized text before any split is drawn. The merged data contain 135,575 rows before exact-text deduplication and 94,323 retained comments afterward, so 41,252 rows are removed. BanTH contracts the most because it carried many near-identical scrapes, which is why its retained share is smaller than its raw size suggests. When records in one group carry conflicting labels, the fixed priority rule of Eq. (1) assigns one final class, the binary flag is set if any record in the group was flagged harmful, and one representative record is retained. That record keeps the source and script of the first row in the deterministic merge order, which the benchmark treats as a primary-source field rather than an assertion about where the comment originated. Grouping before splitting prevents exact normalized duplicates from crossing partitions, and an identifier-intersection assertion after every split confirms it. Fuzzy paraphrase-level duplicates are not detected (Section 7.2). Across the merged data, 1,287 groups span more than one source, covering 3,308 rows. Of those groups, 661 carry conflicting five-class labels over 1,721 rows, and 271 carry conflicting binary flags over 767 rows.

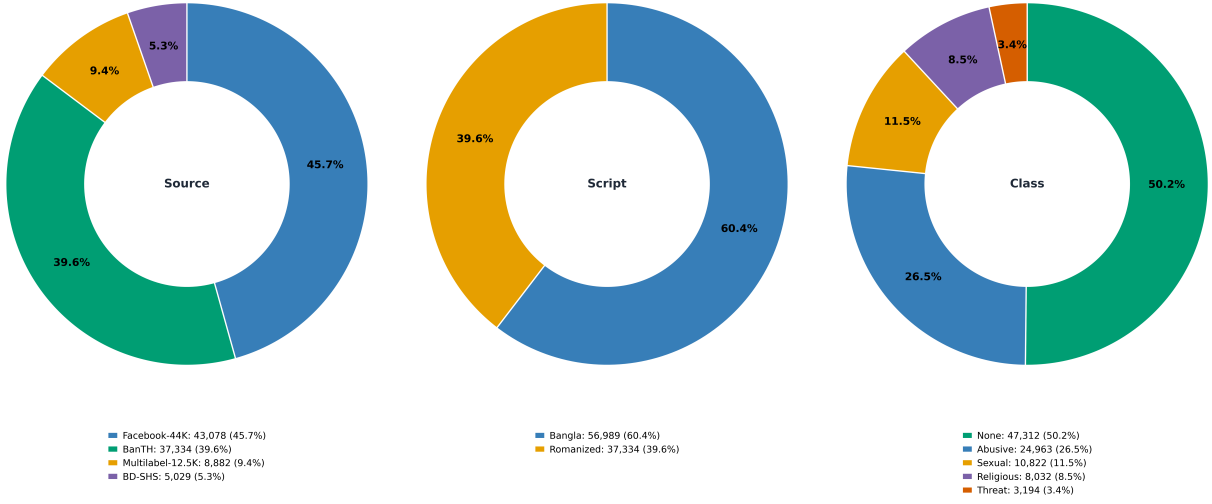


Figure 2. Composition of BanglaCyberBench after deduplication ($n = 94,323$). Left: the four merged sources. Middle: the split between Bangla script (60.4%) and Romanized Bangla (39.6%). Right: the five-class distribution, dominated by *none* and *abusive*, with *threat* at 3.4%.

3.4 The Five-Class Taxonomy

The four sources together use 89 distinct raw label strings, many of them comma- or underscore-compounded (a single comment might be tagged `sexual_threat` or `religious_abusive`). We split each raw string on `,` and `_`, map each piece to a bucket, and resolve compounds by a fixed priority:

$$\text{threat} > \text{sexual} > \text{religious} > \text{abusive} > \text{none}. \quad (1)$$

Explicit violence and identity-targeted abuse take precedence over the generic *abusive* bucket, and *none* loses to every other class. This ordering is a benchmark-construction decision motivated by moderation urgency, not a universal semantic hierarchy, and the released code re-consolidates the data under an alternative rule. Harmonising source taxonomies is necessary because abusive-language datasets frequently use incompatible task and label definitions [38]. Two consistency rules run alongside the priority order. An unrecognised component under a record flagged abusive is sent to *abusive* rather than dropped, and the binary flag has the final say, so a record flagged non-abusive resolves to *none* and one flagged abusive but resolving to *none* is moved to *abusive*. The resulting distribution (Table 3) is imbalanced, with *none* at half the data and *threat* at 3.4%, a ratio of roughly 14.8:1 between the most and least frequent class.

Table 3. Class distribution of the five-class benchmark after deduplication ($n = 94,323$). Raw labels folded into each class are listed in the text.

Class	Samples	Share (%)
None	47,312	50.16
Abusive	24,963	26.47
Sexual	10,822	11.47
Religious	8,032	8.52
Threat	3,194	3.39
Total	94,323	100.00

The five buckets absorb the raw strings as follows. *None* folds in `none` and `not-bully`. *Abusive* is the residual bucket for general offence, covering abusive or violent content, troll, personal offence, body-shaming, origin-based slurs, slander, spam, political abuse, and miscellaneous. *Sexual* folds sexual and gender-based harassment, *Religious* folds religion-directed abuse, and *Threat* folds explicit threats and calls to violence. Table 5 gives the full map.

The five labels do not sit on a single axis. *None* is a harmfulness decision, *threat* marks an act, *sexual* and *religious* are partly target-identity categories, and *abusive* is a residual content bucket. Some folds are arguable, in particular gender-based abuse under *sexual* and origin-based slurs under the generic

abusive bucket. A multi-label or hierarchical representation would capture such cases more faithfully than a flat forced choice. We keep the flat rule for comparability with the source datasets and the comparator, and we do not claim it is the uniquely correct one. A sensitivity study across consolidation policies is left to future work (Section 7.2).

Two authors independently labelled a stratified sample of 500 comments using the five-class taxonomy. The sample was drawn with a fixed seed, stratified jointly by class and script, and labelled blind to the consolidated label and to each other. They agreed on 80.0% of the items. Cohen’s κ [39] was 0.7094, with a 95% bootstrap confidence interval of [0.6583, 0.7579] based on 5,000 resamples [40] (Table 4). The sample contains no Romanized Threat stratum because that combination is absent from the benchmark. Disagreements were left unadjudicated by design, since the aim was to measure convergence on the consolidated scheme rather than to build a corrected gold set. A larger study with more annotators and adjudication is future work (Section 7.2).

Table 4. Agreement between two author-annotators on a stratified sample of 500 comments.

Measure	Value
Observed agreement	80.0%
Cohen’s κ	0.7094
95% bootstrap CI	[0.6583, 0.7579]
Bootstrap resamples	5,000
Sample size	500

A nine-class variant that splits the residual bucket further, adding **personal**, **political**, **gender**, and **other**, is built only for the taxonomy sensitivity analysis of Section 6.4 and never for the headline result.

Table 5. Raw-label consolidation map. Each raw label string is lowercased and split on `,` and `_`; every component is sent to a bucket, and compounds are resolved by the priority order in Eq. (1).

Consolidated class	Raw components folded in
None	none, not bully, notbully
Abusive	abusive, abusive/violence, violence, troll, personal, personal offense, political, slander, spam, origin, body shaming, misc
Sexual	sexual, gender
Religious	religious, religion
Threat	threat, calltoviolence

3.5 Splits and Evaluation Protocol

We report two protocols. The in-domain protocol uses a group-stratified 70/10/20 split of the combined benchmark, giving 66,026 training, 9,432 validation, and 18,865 test comments (Table 6). Stratification holds the five-class proportions fixed across the three partitions, so the rare *threat* and *religious* classes keep the same share in each. The validation split is used for early stopping, model selection within a run, the fusion weights of Eq. (10), and any inference-time tuning. No hyperparameter, checkpoint, or fusion weight of the final model is chosen using test data. The random split measures generalisation to unseen comments from the same consolidated mixture, not generalisation to a new platform. The component and routing analyses of Sections 6.4 and 6.5 also report test-split scores so that they use the same measurement as the headline. The proposed configuration was fixed before those tables were assembled, and they should be read as reported measurements rather than as the selection procedure.

The source-held-out protocol evaluates each of the three Bangla-script sources as the complete test set while splitting the other two into training and validation sets by a stratified 87.5/12.5 partition. BanTH is not included in this protocol because it supplies the benchmark’s Romanized data, and it remains part of the in-domain benchmark. Holding BanTH out would remove every Romanized comment from training and test only on Romanized text, so the resulting score would confound source shift with script shift. Separating the two effects requires a second Romanized source, which we identify as the benchmark’s most useful extension (Section 7.2).

Two further properties of the protocol are worth stating. The script tag $s(x)$ used throughout the

paper is inherited from source provenance rather than detected per comment, which is exact here because each source is script-homogeneous. Because global deduplication attributes text shared across sources to a single primary source, a held-out test set contains the benchmark rows tagged with that source rather than a from-scratch reconstruction of the source as originally collected. The 1,287 groups that span multiple sources therefore sit in training under whichever source they were first attributed to. Of those, 661 carry conflicting five-class labels, so the net effect on transfer difficulty is not signed in one direction. The two split manifests correspond to the in-domain and Bangla-script source-held-out protocols, respectively, and every configuration re-runs the identifier-intersection assertion, so no retained comment appears in more than one partition.

Table 6. Stratified random 70/10/20 split of the benchmark. Stratification is on the five-class label.

Split	Samples	Share (%)
Train	66,026	70.00
Validation	9,432	10.00
Test	18,865	20.00

4. Proposed Model

The proposed model combines four transformer encoders using script-dependent activation and validation-selected logit fusion. Each encoder uses the same pooling head and is trained with cross-entropy and FGM. BanglaBERT is active only for Bangla-script comments, and the other three encoders process both script subsets. This section follows a single comment through the system: tokenisation and encoding (Section 4.1), the pooling and classification head (Section 4.2), the training objective (Section 4.3), the script-dependent activation rule (Section 4.4), and the weighted-logit fusion that produces the final label (Section 4.5). Fig. 3 shows the assembled system, and Table 7 fixes notation.

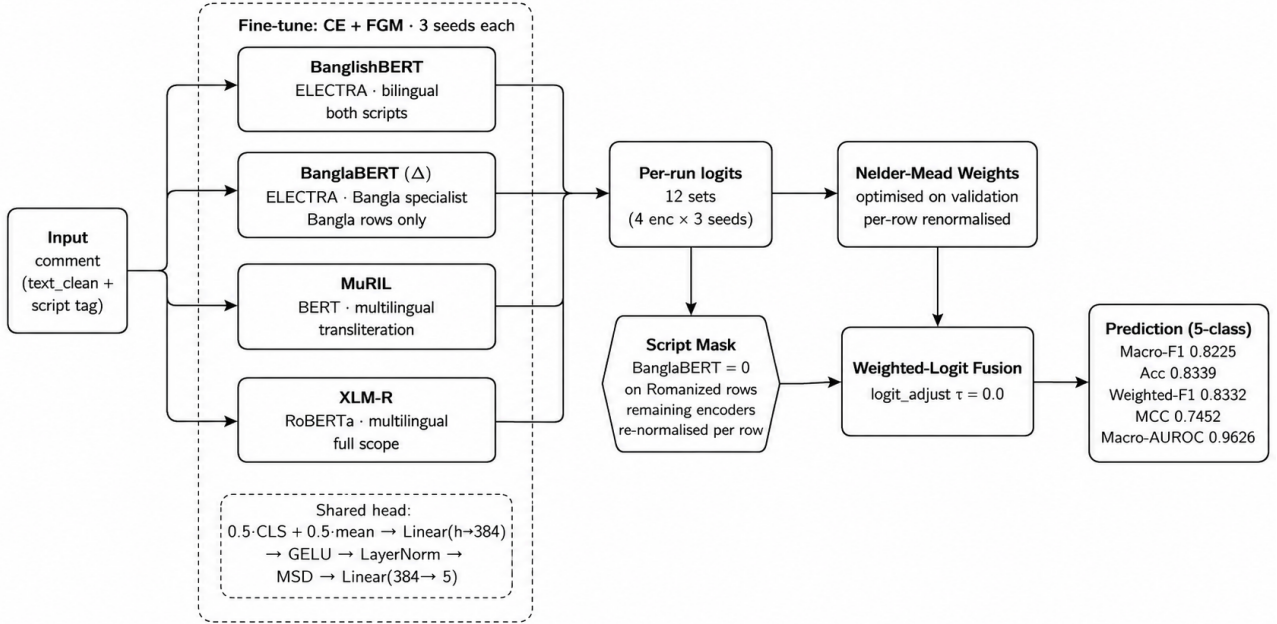


Figure 3. The proposed script-aware ensemble. All four encoders process Bangla-script comments. On Romanized comments BanglaBERT emits neutral logits and receives zero fusion weight, so the remaining encoders decide the label. Each encoder is fine-tuned with cross-entropy and FGM across three seeds, and the twelve resulting logit vectors are fused by validation-selected weights.

4.1 From a Comment to Contextual Embeddings

Each tokenizer truncates or pads a comment to $T \leq 128$ subword tokens, with a leading [CLS] and a trailing [SEP]. The corresponding unmodified 12-layer transformer backbone [41] produces contextual hidden states $(\mathbf{h}_0, \dots, \mathbf{h}_{T-1})$ of width $d = 768$, whose parameters we write as the encoder set θ_m . All

Table 7. Notation used in Section 4.

Symbol	Meaning
x, y	Input comment (token sequence) and its ground-truth class
T, C	Sequence length (≤ 128 tokens) and number of classes ($C = 5$)
$s(x) \in \{\text{BN}, \text{ROM}\}$	Script of comment x : Bangla or Romanized
N_b, N_{test}	Minibatch size and number of test comments
θ_m, ϕ_m	Encoder and head parameters of backbone m
$\mathbf{h}_t \in \mathbb{R}^d$	Contextual hidden state at position t , of width $d = 768$; $\mathbf{h}_{[\text{CLS}]} \equiv \mathbf{h}_0$
$a_t \in \{0, 1\}$	Attention-mask entry indicating a non-padding position
$\alpha_m(x) \in \{0, 1\}$	Activation indicator of member m on comment x
$\mathbf{z}_m \in \mathbb{R}^C$	Logit vector from backbone m
$\hat{p}(y x) \in \Delta^{C-1}$	Predicted class-probability vector
$\mathbf{E}_m, \mathbf{r}$	Token-embedding parameter of member m and its temporary FGM perturbation
ϵ	FGM perturbation budget
$w_m \geq 0$	Fusion weight of ensemble member m , $\sum_m w_m = 1$
$\mathcal{M}, \mathcal{M}(x)$	Full member set (12) and the active members for comment x

backbone parameters are fine-tuned jointly with the pooling and classification head described below. Two of the backbones, BanglaBERT and BanglishBERT [24], are pretrained as ELECTRA discriminators [42] rather than as masked language models. MuRIL and XLM-R are masked-language-model encoders. The pretraining objective matters here only through the representations it produces, and the fine-tuning path below is identical across all four.

4.2 Pooling and Classification Head

Each backbone m carries the same head. The sequence is pooled by an equal-weight blend of the [CLS] vector and a mean over token states:

$$\mathbf{h}_{\text{pool}} = \frac{1}{2} \mathbf{h}_{[\text{CLS}]} + \frac{1}{2} \cdot \frac{\sum_{t=1}^T a_t \mathbf{h}_t}{\sum_{t=1}^T a_t}, \quad (2)$$

where $a_t \in \{0, 1\}$ is the attention-mask entry that zeroes out padding, and m indexes an ensemble member. The pooled vector passes through a small projection with GELU and layer normalisation, then a linear layer to C logits:

$$\mathbf{u} = \text{LN}(\text{GELU}(\mathbf{W}_1 \mathbf{h}_{\text{pool}} + \mathbf{b}_1)) \in \mathbb{R}^{384}, \quad \mathbf{z}_m = \mathbf{W}_2 \mathbf{u} + \mathbf{b}_2 \in \mathbb{R}^C, \quad (3)$$

with $\mathbf{W}_1 \in \mathbb{R}^{384 \times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times 384}$. The class probabilities are $\hat{p}(y | x) = \text{softmax}(\mathbf{z}_m)$. The equal-weight pooling blend and the label-smoothing value defined below are fixed configuration choices. They were not optimized and are not evaluated in a separate ablation.

4.3 Training Objective: Cross-Entropy with FGM

Each backbone is fine-tuned end to end, over both θ_m and ϕ_m , with AdamW [43] under label-smoothed cross-entropy [44]. With smoothing $s = 0.03$ the soft target is $\tilde{y}_c = (1 - s)\mathbb{I}[c = y] + s/C$, and

$$\mathcal{L}_{\text{CE}}(\theta_m, \phi_m) = -\frac{1}{N_b} \sum_{i=1}^{N_b} \sum_{c=1}^C \tilde{y}_{i,c} \log \hat{p}(y = c | x_i). \quad (4)$$

Adversarial training adds a second pass in which the input embeddings are perturbed in the direction that most increases the loss. Written as a min-max problem,

$$\min_{\theta_m, \phi_m} \mathbb{E}_{(x,y)} \left[\max_{\|\mathbf{r}\|_F \leq \epsilon} \mathcal{L}_{\text{CE}}(\mathbf{E}_m + \mathbf{r}; x, y) \right]. \quad (5)$$

FGM [28] is implemented here as a temporary perturbation of the trainable token-embedding parameter, rather than as a persistent change to the input ids. Let $\mathbf{E}_m$ denote the token-embedding parameter of backbone m and let $\mathbf{g}_E = \nabla_{\mathbf{E}_m} \mathcal{L}_{\text{CE}}(\mathbf{E}_m, y)$ be its clean-pass gradient. A first-order Taylor expansion gives $\mathcal{L}_{\text{CE}}(\mathbf{E}_m + \mathbf{r}, y) \approx \mathcal{L}_{\text{CE}}(\mathbf{E}_m, y) + \langle \mathbf{r}, \mathbf{g}_E \rangle_F$, so the inner problem under a Frobenius-norm budget

is maximised by

$$\mathbf{r}_{\text{adv}} = \epsilon \frac{\mathbf{g}_E}{\|\mathbf{g}_E\|_F} \quad (6)$$

a single ascent step on the embedding-parameter ball. The perturbation is applied in-place to the embedding weight for the adversarial forward pass and then restored, so the optimiser never treats it as a new parameter. The per-batch objective sums the clean and adversarial forward passes,

$$\mathcal{L}(\theta_m, \phi_m) = \mathcal{L}_{\text{CE}}(\mathbf{E}_m; x, y) + \mathcal{L}_{\text{CE}}(\mathbf{E}_m + \mathbf{r}_{\text{adv}}; x, y), \quad (7)$$

and both passes accumulate gradients into the same parameters before a single optimiser step (Algorithm 1). FGM costs one extra forward-backward pass per batch and exposes a single hyperparameter ϵ , fixed at 1.0.

Parameter updates. The encoder and classification head are fine-tuned jointly with AdamW [43]. For each minibatch, gradients from the clean and adversarial passes are accumulated before one optimizer update. The encoder and head use separate learning rates (Table 8), gradients are clipped in ℓ_2 norm before the update, and the learning rate is warmed up linearly and then decayed. Section 6.4 reports the component sensitivity analysis that motivates using FGM without additional regularisation.

Algorithm 1 Cross-entropy + FGM fine-tuning of one backbone (single mini-batch).

Require: Batch (x, y) , encoder θ_m , head ϕ_m , budget ϵ

- 1: Encode the batch x with backbone m
 - 2: **Clean pass:** $\mathcal{L}_{\text{clean}} \leftarrow \mathcal{L}_{\text{CE}}(\mathbf{e}, y)$; **backward** (accumulate $\nabla_{\theta_m, \phi_m}$)
 - 3: Read $\mathbf{g}_E = \nabla_{\mathbf{E}_m} \mathcal{L}_{\text{clean}}$; set $\mathbf{r}_{\text{adv}} \leftarrow \epsilon \mathbf{g}_E / \|\mathbf{g}_E\|_F$ ▷ Eq. (6)
 - 4: Temporarily set $\mathbf{E}_m \leftarrow \mathbf{E}_m + \mathbf{r}_{\text{adv}}$
 - 5: **FGM pass:** $\mathcal{L}_{\text{adv}} \leftarrow \mathcal{L}_{\text{CE}}(x, y)$; **backward** (accumulate); **restore** $\mathbf{E}_m$
 - 6: Clip gradients; **optimiser step:** $\theta_m, \phi_m \leftarrow$ AdamW update with configured encoder/head learning-rate groups and decoupled decay
-

4.4 Script-Aware Specialisation

BanglaBERT is excluded from Romanized inputs because its pretraining is primarily Bangla script. BanglaBERT is a Bangla-English bilingual encoder, MuRIL includes transliterated Indic text in pre-training, and XLM-R is multilingual, so those three remain active on both script subsets. BanglaBERT trains, validates, and is evaluated only on Bangla-script rows. On a Romanized comment it emits a neutral zero logit vector and receives zero fusion weight. This routing rule is fixed before test evaluation. It should be interpreted as a pretraining-scope constraint rather than a statistically demonstrated improvement.

Concretely, define an activation indicator $\alpha_m(x)$ for member m on comment x :

$$\alpha_m(x) = \begin{cases} 0 & \text{if } m \text{ is a BanglaBERT member and } s(x) = \text{ROM}, \\ 1 & \text{otherwise.} \end{cases} \quad (8)$$

The active set is $\mathcal{M}(x) = \{m \in \mathcal{M} : \alpha_m(x) = 1\}$. For a Bangla-script comment all twelve members are active, and for a Romanized comment the nine members of the three script-capable backbones carry the decision.

Script metadata are inherited from source because the four source datasets are script-homogeneous under the benchmark protocol (Section 3.5). These fixed metadata determine the routing rule used in the reported experiments. Script and source are therefore perfectly correlated in this benchmark, so Eq. (8) is exact but is effectively source-derived metadata rather than a per-comment detection result.

4.5 Weighted-Logit Fusion

The ensemble has $|\mathcal{M}| = 12$ members, one for each of the four backbones under each of three seeds. Fusion is a script-masked weighted average of logits, renormalised per comment over the active members

so that excluding BanglaBERT on Romanized rows does not shrink the total weight:

$$\mathbf{z}_{\text{ens}}(x) = \sum_{m \in \mathcal{M}(x)} \frac{w_m}{\sum_{m' \in \mathcal{M}(x)} w_{m'}} \mathbf{z}_m(x), \quad \hat{y}(x) = \arg \max_c [\mathbf{z}_{\text{ens}}(x)]_c. \quad (9)$$

The weights are the solution of a constrained maximisation of validation Macro-F1 over the probability simplex,

$$\mathbf{w}^* = \arg \max_{\mathbf{w} \in \Delta^{11}} \text{F1}_{\text{macro}}^{\text{val}}(\mathbf{w}), \quad \Delta^{11} = \left\{ \mathbf{w} \in \mathbb{R}^{12} : w_m \geq 0, \sum_m w_m = 1 \right\}, \quad (10)$$

where $\text{F1}_{\text{macro}}^{\text{val}}(\mathbf{w})$ is the Macro-F1 obtained on the validation split when Eq. (9) is decoded with weights $\mathbf{w}$. The objective is piecewise constant in $\mathbf{w}$, so we solve it with the derivative-free Nelder–Mead simplex search [45], starting from the uniform vector $w_m = \frac{1}{12}$ and clipping candidates back onto Δ^{11} . We select non-negative weights on the validation set under this simplex constraint and fix them before test evaluation. Because uniform and optimized fusion differ by only 0.0003 Macro-F1 in the retained sensitivity analysis (Section 6.5), the exact weight values should not be interpreted as evidence of member importance or complementarity. An optional inference-time logit adjustment for the long tail [35] is available but tuned to zero strength on this benchmark, so the fused logits are the final decision.

5. Experimental Setup

5.1 Backbones and Training Configuration

The experiments use BanglishBERT, BanglaBERT, MuRIL, and XLM-R. All backbones use the same pooling and classification head (Section 4.2). Table 8 lists the training configuration, which is held fixed across backbones so that any difference between them reflects the encoder rather than incidental tuning. Learning rates are uniform across encoder layers, with decoupled AdamW weight decay applied to the configured parameter groups, which is distinct from layer-wise learning-rate decay. The effective batch size is 32. All runs use fp16 mixed precision on a single NVIDIA A6000 (48 GB) with 32 GB of system RAM. Training uses PyTorch [46] and the Transformers library [47], with scikit-learn [48] for scoring and three fixed seeds {42, 123, 456}. Exact package versions and model identifiers belong in the reproducibility repository rather than the main manuscript.

Table 8. Training configuration for the four backbones. Held constant across encoders.

Hyperparameter	Value	Notes
Optimiser	AdamW	$\beta_1 = 0.9, \beta_2 = 0.999$
Encoder / head learning rate	$2 \times 10^{-5} / 8 \times 10^{-5}$	uniform across layers, no layer-wise decay
AdamW weight decay λ	0.01	decoupled, excludes bias and LayerNorm
LR schedule	linear	10% warm-up, then linear decay
Gradient clipping	1.0	global ℓ_2 norm
Model selection	best validation Macro-F1	checkpoint restored before test
Max sequence length	128	covers nearly the whole length distribution
Batch size (effective)	32	physical $32 \times$ accumulation 1
Epochs	up to 8	early stopping, patience 3
Label smoothing	0.03	mild
Dropout / head hidden	0.25 / 384	NA
FGM budget ϵ	1.0	embedding-space perturbation
Precision	fp16	A6000 48 GB
Seeds	42, 123, 456	three per backbone

5.2 Evaluation Metrics

The primary metric is Macro-F1, reported with Weighted-F1 alongside. For C classes,

$$\text{F1}(c) = \frac{2 \text{Prec}(c) \text{Rec}(c)}{\text{Prec}(c) + \text{Rec}(c)}, \quad \text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \text{F1}(c). \quad (11)$$

Macro-F1 weights every class equally, so the rare *threat* and *religious* classes are not hidden by the dominant *none* class. Weighted-F1, which weights by support, is reported for continuity with prior

work. We also report accuracy and the Matthews correlation coefficient [49], which is informative under imbalance because it uses every cell of the confusion matrix $\mathbf{K} \in \mathbb{N}^{C \times C}$. In its multiclass form,

$$\text{MCC} = \frac{\sum_c K_{cc} \cdot N_{\text{test}} - \sum_c \hat{n}_c n_c}{\sqrt{(N_{\text{test}}^2 - \sum_c \hat{n}_c^2)(N_{\text{test}}^2 - \sum_c n_c^2)}}, \quad (12)$$

where N_{test} is the number of test comments, $n_c = \sum_j K_{cj}$ the true count of class c , and $\hat{n}_c = \sum_i K_{ic}$ its predicted count. The coefficient is 1 for perfect prediction and 0 at chance level, and it is not inflated by majority-class dominance in the way accuracy is. We also report one-vs-rest Macro-AUROC, the mean over classes of the area under the ROC curve treating each class against the rest, which measures ranking quality independently of any decision threshold.

5.3 Statistical Reporting

Because the in-domain test split is a single fixed sample, we quantify its uncertainty with a percentile bootstrap [50]. For a metric T we draw $B = 2,000$ resamples of the 18,865 test predictions with replacement, recompute T on each, and report the interval between the 2.5th and 97.5th percentiles:

$$\text{CI}_{95\%} = [\hat{T}^{([0.025B])}, \hat{T}^{([0.975B])}]. \quad (13)$$

The resampling seed is fixed at 42. These intervals are conditional on one fixed test split and one trained final ensemble. They capture test-set sampling noise but not training-seed, source, annotation, or split-construction variance, so they should be read as lower bounds on total uncertainty rather than as a full error budget. We report interval estimates rather than paired significance tests, because paired per-example predictions were not retained for the baselines.

5.4 Computational Cost

BanglaBERT and BanglishBERT contain approximately 110M parameters each, while MuRIL and XLM-R contain approximately 236M and 278M parameters. Each backbone was trained for approximately 35 to 50 minutes on one NVIDIA A6000 GPU at sequence length 128 and batch size 32, so the twelve fine-tuning runs dominate the cost. The fusion step operates on retained model outputs and adds little training cost. FGM doubles the per-batch backward cost during training only. At inference the model performs twelve forward passes, or nine on Romanized input, followed by a weighted average. A single-model alternative is available in our own tables: the strongest full-scope single run reaches 0.8084 Macro-F1 on the same split (Table 11), which is 0.0141 below the ensemble at a twelvefold reduction in inference cost. That alternative must be full-scope, because BanglaBERT abstains on Romanized input by design.

5.5 Component Ablation Design

The proposed configuration fine-tunes with cross-entropy and FGM. The component analysis measures how sensitive Macro-F1 is to that choice and to the imbalance and regularisation components commonly added on top of it. We anchor on BanglishBERT and take plain cross-entropy as the initial reference. FGM is added first, and each subsequent configuration adds one further component to cross-entropy with FGM. Every configuration is run over three seeds, so each value is a mean with a standard deviation and the small differences can be judged against seed variation (Section 6.4). The candidate components are defined below.

Class-balanced focal loss. Focal loss [30] down-weights easy, confident examples so the optimiser attends to hard ones, and class-balanced weighting [31] scales each class by the inverse of its effective sample count. With focusing parameter γ and per-class weight ω_c ,

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N_b} \sum_{i=1}^{N_b} \omega_{y_i} (1 - \hat{p}_{i,y_i})^\gamma \log \hat{p}_{i,y_i}, \quad \omega_c \propto \frac{1 - \beta}{1 - \beta^{n_c}}, \quad (14)$$

where n_c is the count of class c and $\beta \rightarrow 1$ recovers inverse-frequency weighting. We use $\gamma = 2$ and $\beta = 0.999$.

Balanced sampler. Instead of reweighting the loss, this reshapes the batch: each example is sampled with probability proportional to $n_{y_i}^{-1/2}$ (square-root inverse frequency), so minority classes appear more

often during training.

Multi-sample dropout. The head is evaluated under K independent dropout masks and the losses are averaged, which reduces the variance of the head update at no inference cost [32]:

$$\mathcal{L}_{\text{MSD}} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CE}}(\text{head}(\text{dropout}_k(\mathbf{h}_{\text{pool}}))). \quad (15)$$

R-Drop. Two independent dropout passes over the same input give two distributions $\hat{p}^{(1)}$ and $\hat{p}^{(2)}$. R-Drop [33] adds their symmetric KL divergence to the loss, which pushes the two passes to agree:

$$\mathcal{L}_{\text{RDrop}} = \mathcal{L}_{\text{CE}} + \frac{\lambda}{2} [\text{KL}(\hat{p}^{(1)} \parallel \hat{p}^{(2)}) + \text{KL}(\hat{p}^{(2)} \parallel \hat{p}^{(1)})], \quad (16)$$

with $\lambda = 0.5$.

Exponential moving average (EMA). A shadow copy of the weights is maintained as an exponential moving average, $\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu) \theta$ with $\mu = 0.999$, and used for evaluation. It is retained only when it obtains a higher validation Macro-F1 than the raw weights.

The full-stack configuration enables all of these components at once.

6. Results and Analysis

We report in-domain benchmark results with bootstrap confidence intervals (Section 6.1), a zero-shot large-language-model reference (Section 6.2), a per-class and error breakdown (Section 6.3), the component and taxonomy sensitivity analyses (Section 6.4), script-stratified performance and routing sensitivity (Section 6.5), source-held-out generalisation (Section 6.6), the protocol-aligned comparison with the published comparator (Section 6.7), and the selected fusion weights (Section 6.8).

6.1 In-Domain Benchmark

On the in-domain test split ($n = 18,865$), the ensemble achieves a Macro-F1 of 0.8225 with a 95% bootstrap confidence interval of [0.8146, 0.8300]. Weighted-F1 is 0.8332, accuracy is 0.8339, and MCC is 0.7452 (Tables 9 and 10). Macro-AUROC is 0.9626. These intervals quantify uncertainty for the ensemble’s own predictions. They do not establish statistical superiority over the baselines, because paired baseline predictions are unavailable. Among the retained non-transformer and neural baselines in Table 23, the word-and-character TF-IDF linear SVM is strongest, with a Macro-F1 of 0.7674. The comparison with that baseline is not data-budget matched: the classical baselines are fitted on train and validation (80%) and need no checkpoint selection, whereas the transformers optimise on train (70%) and use validation for early stopping. Fig. 4a shows the comparison, and Fig. 4b shows the intervals.

Table 9. Main results on the in-domain test split ($n = 18,865$). The reference is the strongest retained non-transformer baseline, a TF-IDF word and character linear SVM. The full baseline set is given in Table 23.

System	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
Best non-transformer baseline	0.7674	0.7889	0.7933	0.6788	0.9418
Proposed model	0.8225	0.8332	0.8339	0.7452	0.9626

Table 10. Percentile-bootstrap 95% confidence intervals for the proposed model on the in-domain test split ($B = 2,000$, generator seed 42). Widths are computed from the rounded endpoints shown.

Metric	Value	95% CI	CI width
Macro-F1	0.8225	[0.8146, 0.8300]	0.0154
Weighted-F1	0.8332	[0.8276, 0.8386]	0.0110
Accuracy	0.8339	[0.8285, 0.8393]	0.0108
MCC	0.7452	[0.7369, 0.7536]	0.0167

Per-backbone single-run scores (Table 11) also serve as the transformer baseline suite, since each row

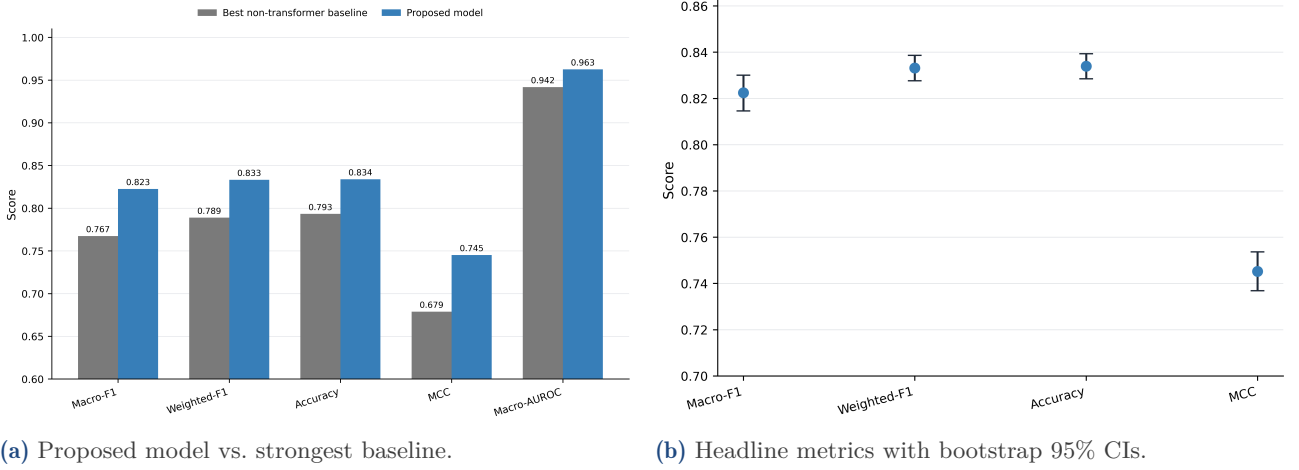


Figure 4. In-domain test performance. The proposed model reports a higher point estimate than the strongest retained non-transformer baseline on every metric, and the bootstrap intervals are narrow (Macro-F1 width 0.0154).

is a pretrained encoder fine-tuned with the identical configuration. The table spans two evaluation scopes and keeps them apart. BanglishBERT, MuRIL, and XLM-R are full-scope models scored on the complete test split, and their best single run is MuRIL at 0.8084, with BanglishBERT at 0.8083 and XLM-R at 0.8052 close behind. BanglaBERT is restricted to Bangla-script rows by the routing rule, so its 0.8154 is measured on the Bangla-script subset alone and cannot be lined up against the full-scope rows. The ensemble’s Macro-F1 is 0.8225, compared with 0.8084 for the strongest retained full-scope member. This is an unpaired point-estimate difference, because the required paired member predictions were not retained for the final configuration. The result should not be described as statistically significant or as proof of complementary errors.

Table 11. Per-backbone, per-seed single-run scores, each trained with the identical configuration. BanglishBERT, MuRIL, and XLM-R are full-scope and are scored on the complete test split ($n = 18,865$). BanglaBERT is restricted to Bangla-script rows by the routing rule of Section 4.4, so its rows are scored on the Bangla-script test subset only and are not directly comparable with the full-scope rows.

Backbone	Seed	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
BanglishBERT	42	0.8069	0.8201	0.8206	0.7252	0.9567
BanglishBERT	123	0.7996	0.8124	0.8161	0.7151	0.9547
BanglishBERT	456	0.8083	0.8200	0.8213	0.7261	0.9568
BanglaBERT [†]	42	0.8105	0.8178	0.8177	0.7545	0.9602
BanglaBERT [†]	123	0.8154	0.8211	0.8218	0.7603	0.9599
BanglaBERT [†]	456	0.8149	0.8205	0.8216	0.7600	0.9615
MuRIL	42	0.8055	0.8177	0.8185	0.7231	0.9521
MuRIL	123	0.8084	0.8216	0.8222	0.7284	0.9502
MuRIL	456	0.8066	0.8198	0.8210	0.7259	0.9523
XLM-R	42	0.8052	0.8159	0.8161	0.7199	0.9538
XLM-R	123	0.8016	0.8144	0.8145	0.7166	0.9503
XLM-R	456	0.8035	0.8173	0.8181	0.7215	0.9501

[†]Bangla-script specialist: trained and evaluated on Bangla-script rows only, so these scores are over the Bangla-script test subset and are not comparable with the full-scope rows above. Bold marks the best full-scope run.

6.2 Zero-Shot LLM Reference

We evaluate Gemini 2.5 Flash under one fixed zero-shot prompt [51]. The prompt gave the same five-class definitions and priority rule the annotators used, and the model labelled each of the 18,865 test comments individually at temperature 0 with an instruction to emit the bare label word. Replies were matched against the five class names, and a reply matching none of them fell back to *none*, the majority class. It obtains a Macro-F1 of 0.5597 on the retained test split (Table 12). Because only one model and prompt were evaluated and aggregate parse-failure records were not retained, this result is a limited reference rather than a general comparison of prompted language models.

Table 12. Zero-shot large-language-model reference against the proposed supervised model on the in-domain test split ($n = 18,865$).

System	Macro-F1	Weighted-F1	Accuracy	MCC
Gemini 2.5 Flash (zero-shot)	0.5597	0.6684	0.6647	0.4957
Proposed model (supervised)	0.8225	0.8332	0.8339	0.7452

The supervised ensemble reports a higher point estimate than this single zero-shot configuration. Prompt design, model choice, and parsing behavior were not varied, so no broader conclusion about language-model-based moderation is supported.

6.3 Per-Class Performance and Error Analysis

Table 13 reports per-class performance on the in-domain test split. *Religious* and *none* obtain the highest F1 values, 0.9031 and 0.8771. *Sexual* follows at 0.8238, and the two lowest are *threat* at 0.7689 and *abusive* at 0.7397. *Threat* has the smallest support, 638 test items, and correspondingly the widest bootstrap interval, [0.7424, 0.7949]. *Abusive* is the residual bucket, absorbing troll, personal, political, and spam content, so its boundary with *none* is less sharply defined.

Table 13. Per-class precision, recall, F1, and bootstrap 95% confidence interval on the in-domain test split. Support is the number of test comments in each class.

Class	Precision	Recall	F1	95% CI (F1)	Support
Religious	0.9221	0.8848	0.9031	[0.8921, 0.9137]	1,606
None	0.8622	0.8925	0.8771	[0.8723, 0.8819]	9,463
Sexual	0.8359	0.8120	0.8238	[0.8115, 0.8358]	2,165
Threat	0.7878	0.7508	0.7689	[0.7424, 0.7949]	638
Abusive	0.7532	0.7266	0.7397	[0.7297, 0.7497]	4,993
Macro avg.	0.8322	0.8133	0.8225	[0.8146, 0.8300]	18,865

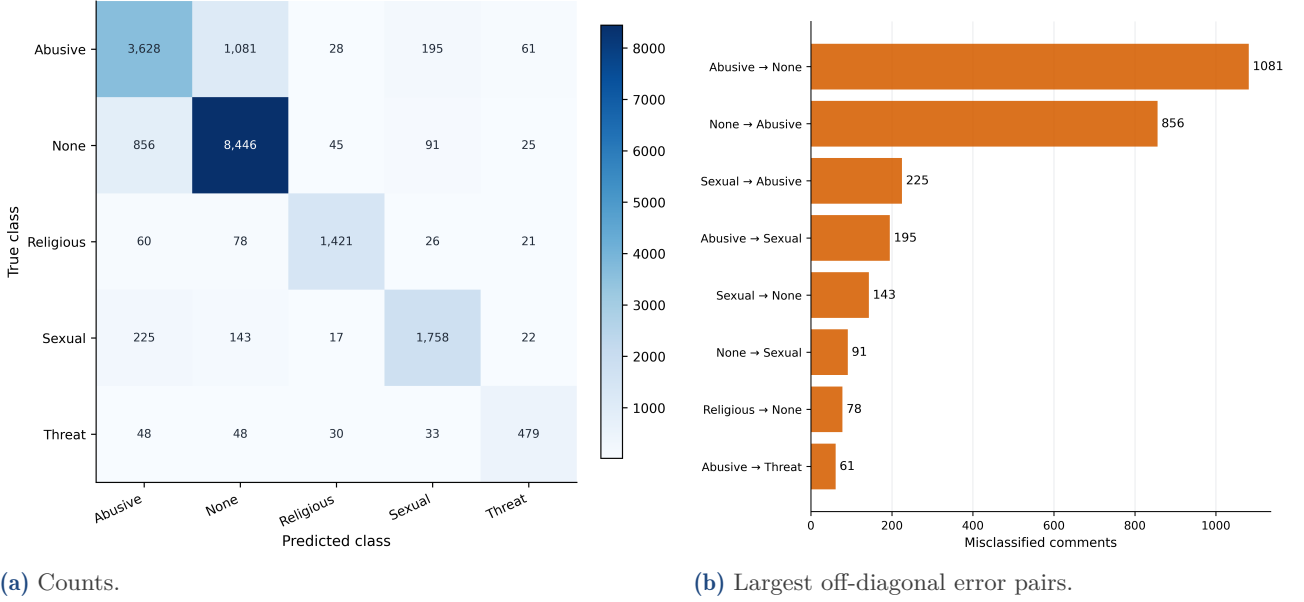
Every class has positive support in the overall test-split results. For script-stratified results, a class with zero gold support contributes an F1 of 0 to the five-class mean. This applies to Romanized Bangla, whose consolidated gold set contains no *threat* examples, so Section 6.5 reports both the fixed five-class value and a supported-class value and labels each explicitly.

The confusion matrix (Fig. 5a) shows where the errors concentrate. The largest off-diagonal mass is the *abusive* and *none* pair, with 1,081 true-*abusive* comments predicted *none* and 856 true-*none* predicted *abusive*. The next-largest confusions (Fig. 5b) are *sexual* predicted as *abusive* (225) and *abusive* predicted as *sexual* (195). *Threat*, despite the smallest support, keeps 479 of its items on the diagonal. The three summarized confusion patterns account for 2,405 of the model’s 3,133 errors, or 76.8%.

Most errors occur between Abusive and None and between Sexual and Abusive (Table 14). These patterns may reflect semantic overlap or differences in annotation policy between the four source teams, but the present study does not include a coded qualitative error analysis. The explanations are therefore hypotheses rather than demonstrated causes. A third pattern, *threat* predicted as *abusive*, is consistent with *threat* recall (0.7508) trailing its precision (0.7878), though a confusion matrix alone cannot establish the mechanism.

Table 14. The three largest off-diagonal error patterns, with a hypothesised mechanism and the number of test comments each accounts for. Together they cover 2,405 of the 3,133 errors.

Error pattern	Mechanism	Count
Abusive ↔ None	Intent ambiguity: sarcasm, in-group language, and blunt-but-clean criticism are not separable from the surface text alone.	1,937
Sexual ↔ Abusive	Content overlap: misogynistic comments are also generically abusive, and the single-label scheme forces one class.	420
Threat → Abusive	Minority dilution: implicit threats without violent keywords drift into the larger <i>abusive</i> bucket.	48



(a) Counts.

(b) Largest off-diagonal error pairs.

Figure 5. Confusion structure of the proposed model on the in-domain test split. The largest off-diagonal counts occur between Abusive and None.

6.4 Ablations

Component sensitivity. Table 15 starts from plain cross-entropy and adds components one at a time, with three seeds per configuration. In this fixed BanglishBERT sensitivity configuration, adding FGM increases Macro-F1 from 0.7961 to 0.8071, a difference of 0.0110 that is larger than the observed seed spread of the reference (0.0022). We state only that FGM increases Macro-F1 by 0.0110 over cross-entropy in this configuration, not that it is universally decisive.

The remaining rows measure each additional component against cross-entropy with FGM. EMA is 0.0022 higher and R-Drop 0.0012 higher, multi-sample dropout is 0.0003 lower, and focal loss with class weighting, the balanced sampler, and the full stack are 0.0016, 0.0076, and 0.0029 lower. Because the configurations differ and each uses only three seeds, we attach no significance claims to these differences. Cross-entropy with FGM was retained as the transparent and computationally modest choice. The sensitivity configuration also differs from the main training configuration in batch size, learning rates, and epoch budget, so these values are not a tuned leaderboard. Fig. 6 plots the same measurements.

Table 15. Component sensitivity on the retained test split using BanglishBERT and three seeds per configuration (mean $\pm$ std). CE is the initial reference. FGM is then added, and each subsequent row adds the named component to CE+FGM. Differences are descriptive rather than significance tests.

Configuration	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC	$\Delta_{\text{CE+FGM}}$
CE only (initial reference)	0.7961 $\pm$ 0.0022	0.8087	0.8093	0.7079	0.9470	NA
CE + FGM (reference)	0.8071 $\pm$ 0.0009	0.8199	0.8218	0.7262	0.9557	0.0000
CE + FGM + focal + class weights	0.8055 $\pm$ 0.0012	0.8178	0.8197	0.7232	0.9553	-0.0016
CE + FGM + balanced sampler	0.7995 $\pm$ 0.0026	0.8115	0.8114	0.7129	0.9428	-0.0076
CE + FGM + multi-sample dropout	0.8068 $\pm$ 0.0014	0.8187	0.8207	0.7248	0.9564	-0.0003
CE + FGM + R-Drop	0.8083 $\pm$ 0.0006	0.8206	0.8231	0.7280	0.9568	+0.0012
CE + FGM + EMA	0.8093 $\pm$ 0.0004	0.8212	0.8220	0.7273	0.9576	+0.0022
Full stack (all on)	0.8042 $\pm$ 0.0019	0.8145	0.8143	0.7176	0.9529	-0.0029

Taxonomy sensitivity. The nine-class sensitivity configuration, which adds **personal**, **political**, **gender**, and **other**, produces a lower Macro-F1 than the five-class configuration, 0.6140 against 0.8073 (Table 16, Fig. 7). Weighted-F1 changes less because the majority classes dominate it. The result is consistent with limited support for several fine-grained classes, but the retained analysis does not establish support as the sole cause.

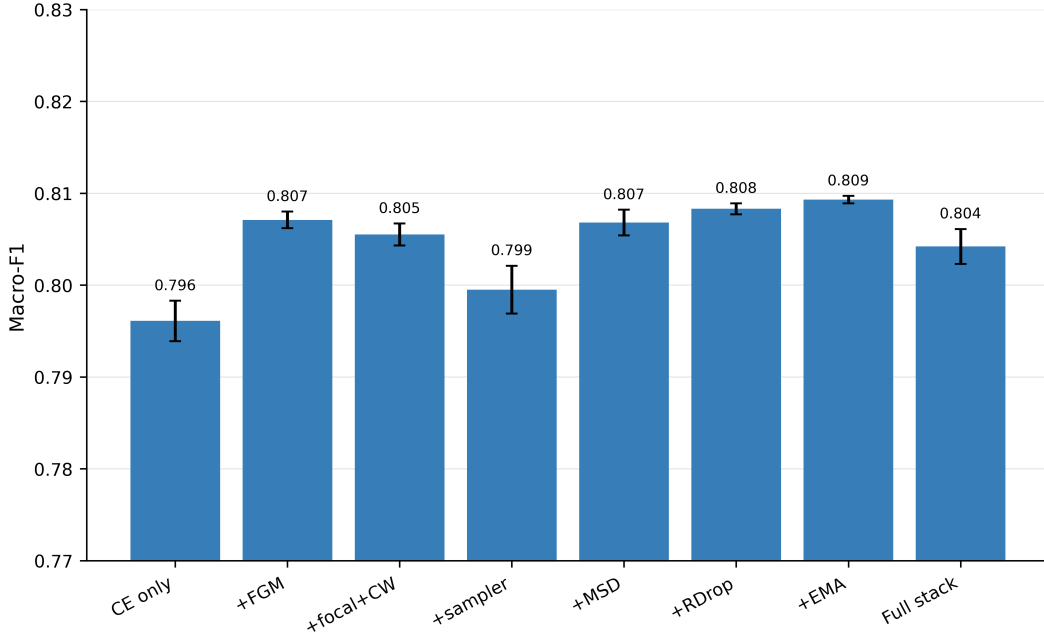


Figure 6. Component sensitivity, mean $\pm$ std over three seeds. FGM produces the largest first increase in this configuration, and the proposed model fine-tunes each backbone with cross-entropy and FGM.

Table 16. Taxonomy sensitivity under the reported training configuration. The nine-class scheme has a substantially lower Macro-F1, while Weighted-F1 changes less.

Taxonomy	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
5-class (headline)	0.8073	0.8203	0.8214	0.7255	0.9556
9-class (ablation)	0.6140	0.8020	0.8069	0.7071	0.9460

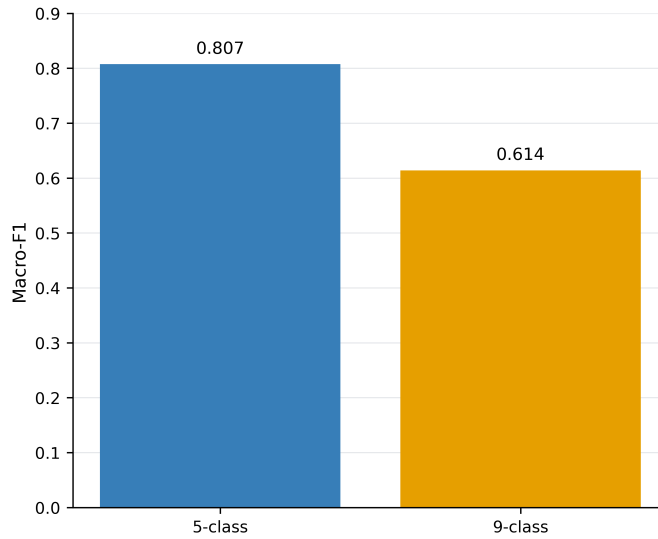


Figure 7. Taxonomy sensitivity. Macro-F1 is 0.6140 for the nine-class configuration and 0.8073 for the five-class configuration.

6.5 Script-Stratified Performance and Routing Sensitivity

Table 17 reports final-ensemble performance by script on the in-domain test split. Romanized performance is reported under both the fixed five-class definition and a supported-class definition. The latter excludes classes with zero gold support in that subset.

The difference between the two Romanized Macro-F1 definitions is caused by zero *threat* support. Religious and Sexual support is also small in the Romanized test subset, so their class-specific estimates are unstable. Table 18 gives the per-class values behind these aggregates.

Because Romanized gold has zero support for *threat*, and that class contributes zero to the five-class

Table 17. Final-ensemble results by script on the in-domain test split.

Subset	n	Macro-F1 ₅	Macro-F1 _{sup}	Weighted-F1	Accuracy	MCC
All	18,865	0.8225	0.8225	0.8332	0.8339	0.7452
Bangla	11,406	0.8303	0.8303	0.8368	0.8369	0.7805
Romanized	7,459	0.3900	0.4875	0.8238	0.8293	0.5771

Table 18. Per-class F1 and support by script on the in-domain test split. Support counts are for the test partition, not the full benchmark.

Class	Bangla F1	Bangla n	Romanized F1	Romanized n
Abusive	0.7727	2,916	0.6917	2,077
None	0.8656	4,162	0.8861	5,301
Religious	0.9104	1,576	0.3721	30
Sexual	0.8340	2,114	0.0000	51
Threat	0.7689	638	NA	0

mean, the five-class Romanized Macro-F1 is capped at $4/5 = 0.80$ even for a classifier that is perfect on the four supported classes. Every Romanized five-class value must be read against that ceiling. The supported-class mean is a separate descriptive quantity and should not be conflated with the fixed five-class value. Table 24 in Section A reports supported-class Macro-AUROC, expected calibration error, and Brier score for the same three subsets. Those are in-domain values and are not source-held-out calibration estimates.

Routing sensitivity. The routing comparison in Table 19 uses a separate cached sensitivity configuration

Table 19. Routing sensitivity on the cached sensitivity configuration. Excluding BanglaBERT from Romanized comments is compared against letting every backbone vote on every comment. A zero-support class contributes an F1 of 0 to the five-class mean.

Setting	Romanized Macro-F1 ₅	Overall Macro-F1
No routing mask (all backbones vote)	0.4568	0.8183
Script-aware routing (proposed)	0.4774	0.8176

In the sensitivity configuration, masking BanglaBERT changes Romanized five-class Macro-F1 from 0.4568 to 0.4774 and changes overall Macro-F1 from 0.8183 to 0.8176. These point estimates come from the sensitivity run, not the final ensemble. The routing mask is retained as a pretraining-scope constraint, but the available outputs do not establish a statistically significant improvement.

Fusion rule. Uniform and validation-optimized fusion obtain Macro-F1 values of 0.8196 and 0.8193 in the same sensitivity configuration, against 0.8073 for the best single backbone in that study (Table 20). Their 0.0003 difference does not support a claim that optimized weighting improves performance.

Table 20. Fusion-rule sensitivity on the cached sensitivity-run outputs. Uniform averaging and the validation-selected Nelder–Mead rule differ by 0.0003 Macro-F1.

Fusion rule	Macro-F1
Best single backbone	0.8073
Uniform average (equal weights)	0.8196
Nelder–Mead weighted (validation-selected)	0.8193

6.6 Cross-Source Robustness

Random-split evaluation measures generalisation to unseen comments from the same mixture. The source-held-out protocol of Section 3.5 measures generalisation to an unseen source. Each of the three Bangla-script sources is held out in turn as the complete test set, the other two are split into training and validation, and the identifier-intersection assertion confirms that no comment appears in two

partitions. Table 21 reports the three configurations with their split sizes, and Fig. 8 plots them against the in-domain value.

Table 21. Source-held-out results and split sizes for the three Bangla-script sources. Macro-F1 is the mean $\pm$ standard deviation over three seeds. The in-domain reference is a Macro-F1 of 0.8225.

Held-out source	n_{train}	n_{val}	n_{test}	Macro-F1	MCC
Facebook-44K	12,172	1,739	43,078	0.5850 ± 0.0075	0.5291
Multilabel-12.5K	42,093	6,014	8,882	0.5601 ± 0.0123	0.4515
BD-SHS	45,465	6,495	5,029	0.4612 ± 0.0039	0.4088

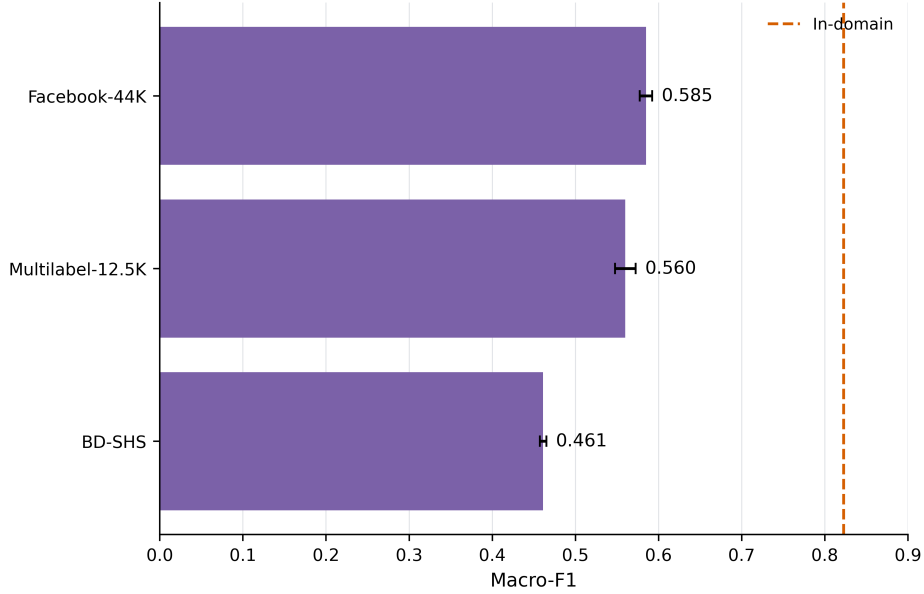


Figure 8. Source-held-out Macro-F1 (mean $\pm$ std over three seeds) against the in-domain reference (dashed).

Source-held-out Macro-F1 ranges from 0.4612 to 0.5850, substantially below the in-domain result. The configuration with the smallest training pool does not have the lowest held-out score, so training size alone does not explain the ordering. The held-out sources also differ in class distribution and annotation policy, and the experiment measures their combined effect rather than attributing the difference to one factor. Table 25 in Section A shows substantial class-distribution differences between the training and held-out sets. These differences are one component of the measured source shift, but the experiment does not isolate their effect from language, sampling, or annotation-policy differences. The seed spread is tight, with standard deviations between 0.0039 and 0.0123, which indicates that the observed shift is reproducible for this protocol rather than universal across Bengali corpora. Macro-AUROC remains between 0.83 and 0.87 while Macro-F1 falls. Source-held-out probabilities were not retained, so we do not report expected calibration error or Brier scores for these configurations and we make no calibration claim from AUROC alone. Target-source validation or threshold recalibration is a plausible mitigation, but its effect is not measured here. The practical implication is that in-domain performance should not be treated as evidence of readiness on an unseen source.

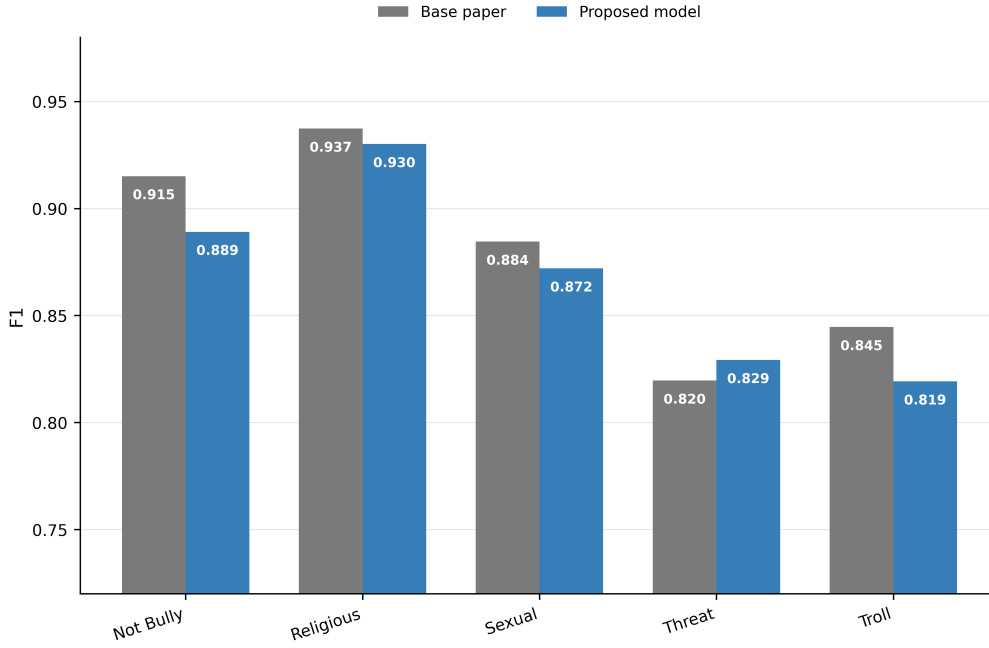
6.7 Comparison with the Base Paper

We retrain the model under the published comparator’s protocol: the Facebook-44K source, within-source deduplication, the native five classes (Not Bully, Sexual, Troll, Religious, Threat), and a 70/15/15 split. The comparator evaluates eight pretrained backbones under an ensemble and MLP stacking protocol and does not use FGM, so our row is a protocol-aligned variant with four backbones, cross-entropy with FGM, and our documented preprocessing, rather than a reimplement of its training procedure. Under this protocol the proposed variant obtains a Macro-F1 of 0.8679, compared with the reported value of 0.8803. Threat F1 is 0.8292 for the proposed variant and 0.8197 in the comparator. The evaluated splits are not identical, so the comparison is descriptive and unpaired (Table 22).

Fig. 9 plots the per-class values. The proposed variant is 0.0124 below the comparator in Macro-F1

Table 22. Protocol-aligned comparison with the published Facebook-44K results [6]. The comparison is unpaired because the evaluated splits are not identical.

Class / Metric	Base paper	Proposed model	Δ
Not Bully	0.9151	0.8891	-0.0260
Religious	0.9374	0.9302	-0.0072
Sexual	0.8845	0.8720	-0.0125
Troll	0.8446	0.8192	-0.0254
Threat	0.8197	0.8292	+0.0095
Macro-F1	0.8803	0.8679	-0.0124
Weighted-F1	0.8923	0.8736	-0.0187
Accuracy	NA	0.8737	NA
MCC	NA	0.8314	NA
Macro-AUROC	NA	0.9747	NA

**Figure 9.** Per-class F1 under the Facebook-44K protocols. Values are descriptive because the evaluated splits are not identical.

and 0.0095 above it in Threat F1. These are unpaired point estimates under protocol-aligned but non-identical data splits. Our copy of the source carries 43,078 comments after within-source deduplication, against the 44,001 reported by the comparator, and the published scores are not predictions on our test indices. The two systems also differ in training configuration, since our variant uses cross-entropy with FGM and the comparator does not. We therefore make no claim of statistical superiority or of a new single-dataset state of the art.

6.8 Ensemble Weights

The selected weights (Fig. 10) are distributed across several members rather than concentrated on one. Optimized and uniform fusion differ by only 0.0003 Macro-F1 in the retained sensitivity analysis (Table 20). Weight magnitude therefore should not be interpreted as direct evidence of member quality, error complementarity, or a gain from optimization.

The weights were fitted on the validation split alone under a single non-nested design, constrained to the simplex, and frozen before the test readout. Fitting eleven free parameters on 9,432 validation examples carries some risk of validation overfitting, but the fusion-rule sensitivity analysis bounds how much that can matter, since uniform averaging is within 0.0003 of the selected rule. We do not isolate the marginal value of each added backbone and seed, which would require a controlled ensemble-design study.

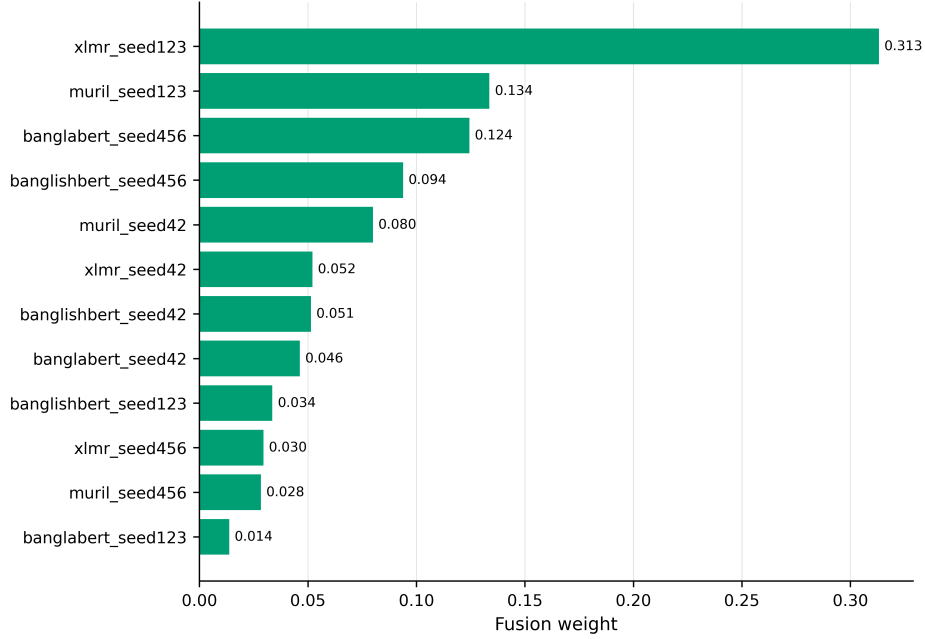


Figure 10. Validation-selected fusion weights for the twelve ensemble members.

7. Discussion

7.1 What the Results Show

In-domain Macro-F1 is 0.8225, while source-held-out Macro-F1 ranges from 0.4612 to 0.5850. FGM produces the largest first point-estimate increase in the component sensitivity study. Script-stratified evaluation shows a lower fixed five-class Macro-F1 for Romanized comments, where Religious and Sexual support is small and Threat support is zero. These results reproduce known cross-dataset degradation in a Bengali setting and show why script-level scores should be reported together with class support.

The routing rule follows from the pretraining scope of the encoders rather than from a measured improvement. BanglaBERT is pretrained primarily on Bangla script, so it is excluded from Romanized inputs, and the bilingual and multilingual encoders remain active on both subsets. The selected fusion weights are distributed across members, but uniform averaging matches the selected rule to within 0.0003 Macro-F1, so the weights are not evidence that individual members contribute different competencies.

Under the comparator’s own Facebook-44K protocol, the proposed variant is 0.0124 below the reported Macro-F1 and 0.0095 above the reported Threat F1. Both are unpaired point estimates on non-identical splits. The proposed system is built for four sources and two scripts rather than for one clean single-source corpus, and we make no claim that either difference is statistically established.

7.2 Limitations and Threats to Validity

Several limitations bound the claims of this study. Script-stratified differences should be interpreted descriptively, because the Romanized comments come from BanTH and the Bangla-script comments come from the other three sources, so script and source are not separable in this benchmark. The Romanized test subset contains only 30 Religious and 51 Sexual examples and no Threat examples, so the corresponding class estimates are unstable and the fixed five-class Romanized Macro-F1 is capped at 0.80 by convention. Paired predictions were not retained for the final member comparison, the routing comparison, or the comparator comparison, so those differences are descriptive rather than tested. Exact-text deduplication does not detect paraphrase-level overlap. Any residual near-duplicate crossing the train and test boundary would inflate the in-domain figure rather than deflate it.

Further limitations concern scope.

These limitations qualify the interpretation of the reported results. They do not invalidate the benchmark construction, and they do not require additional experiments for the present study.

8. Conclusion and Future Work

We evaluate a shared five-class Bengali harmful-language benchmark across Bangla and Romanized scripts. BanglaCyberBench consolidates four public sources into 94,323 comments under a documented taxonomy, and the script-aware ensemble of four encoders is trained with cross-entropy and FGM. The ensemble obtains a Macro-F1 of 0.8225 in-domain with percentile-bootstrap intervals, while source-held-out performance ranges from 0.4612 to 0.5850. Script-stratified results also show lower support for several Romanized minority classes, with no Threat examples at all. These findings support source-held-out and script-stratified reporting with explicit class support for Bengali harmful-language detection.

Future work may retain paired predictions for statistical testing, add further Romanized sources so that script and source effects can be separated, and examine transliteration or cross-script adaptation. A severity-aware or multi-label extension of the taxonomy is also possible once the annotated data can support it. These are future directions rather than requirements for the present study.

Declarations

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing Interests. The authors declare no competing interests.

Ethical Considerations. The study uses previously released research datasets and does not collect new social-media data. Two authors annotated an already collected sample for taxonomy validation. The material contains offensive, sexual, religious, and threatening language, so the release carries a content warning. Dataset redistribution and removal procedures must follow the licences and terms of the original sources. The study did not conduct a comprehensive PII or fairness audit.

Data and Code Availability. Processing code, evaluation code, predictions, split metadata, and permitted research artifacts are available at <https://github.com/Sefayet-Alam/BanglaCyberBench>. Source text remains subject to the licences and redistribution terms of the original datasets.

Author Contributions. S.A. led the conceptualisation, methodology, and implementation, ran all training and evaluation, performed the analysis, and wrote the original draft. S.A. and N.P. carried out the independent label validation and annotation study of Section 3.4. N.P. and A.F.M.M.R. contributed to the experimental design, the interpretation of results, and the review and editing of the manuscript. All authors read and approved the final version.

Declaration of Generative AI and AI-assisted Technologies. The authors used AI-assisted tools only to support software development, debugging, and the scripted generation of result tables and figures. All research design, analysis, interpretation, and manuscript writing were carried out by the authors, who reviewed all outputs and take full responsibility for the content of this publication.

A. Supplementary Tables

Table 23. Retained non-transformer and neural baseline results on the in-domain test split.

System	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
TF-IDF + logistic regression	0.5025	0.6142	0.6025	0.4273	0.8460
TF-IDF word/character n -grams + linear SVM	0.7674	0.7889	0.7933	0.6788	0.9418
BiLSTM	0.6850	0.7106	0.7065	0.5663	0.9052

References

- [1] A. Arango, J. Pérez, and B. Poblete, “Hate speech detection is not as easy as you may think: A closer look at model validation,” in *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2019, pp. 45–54. [Online]. Available: <https://doi.org/10.1145/3331184.3331262>

Table 24. In-domain ranking and calibration measures by script. Supported AUROC excludes classes with zero gold support in the subset. These are in-domain values and are not source-held-out calibration estimates.

Subset	Supported AUROC	ECE	Brier score
All	0.9626	0.0328	0.2482
Bangla	0.9671	0.0348	0.2454
Romanized	0.8743	0.0301	0.2524

Table 25. Class shares in the training and held-out test sets for each source-held-out configuration.

Held-out source	Split	None	Abusive	Religious	Sexual	Threat
Facebook-44K	Train	41.4%	31.3%	3.3%	13.2%	10.8%
Facebook-44K	Test	34.4%	24.1%	17.3%	20.3%	3.9%
Multilabel-12.5K	Train	36.3%	24.1%	15.7%	19.0%	5.0%
Multilabel-12.5K	Test	35.5%	35.4%	4.1%	16.5%	8.6%
BD-SHS	Train	34.6%	26.0%	15.0%	19.7%	4.7%
BD-SHS	Test	51.8%	24.0%	2.0%	7.4%	14.7%

- [2] S. D. Swamy, A. Jamatia, and B. Gambäck, “Studying generalisability across abusive language detection datasets,” in *Proceedings of the 23rd Conference on Computational Natural Language Learning*. Association for Computational Linguistics, 2019, pp. 940–950. [Online]. Available: <https://aclanthology.org/K19-1088/>
- [3] M. Wiegand, J. Ruppenhofer, and T. Kleinbauer, “Detection of abusive language: The problem of biased datasets,” in *Proceedings of NAACL-HLT 2019*. Association for Computational Linguistics, 2019, pp. 602–608. [Online]. Available: <https://aclanthology.org/N19-1060/>
- [4] P. Fortuna, J. Soler-Company, and L. Wanner, “How well do hate speech, toxicity, abusive and offensive language classification models generalize across datasets?” *Information Processing & Management*, vol. 58, no. 3, p. 102524, 2021. [Online]. Available: <https://doi.org/10.1016/j.ipm.2021.102524>
- [5] W. Yin and A. Zubiaga, “Towards generalisable hate speech detection: A review on obstacles and solutions,” *PeerJ Computer Science*, vol. 7, p. e598, 2021. [Online]. Available: <https://doi.org/10.7717/peerj-cs.598>
- [6] M. N. Hoque, R. P. D. Nath, A. N. Chy, D. Ghose, and M. H. Seddiqui, “Advancing cyberbullying detection in low-resource languages: a transformer-stacking framework for Bengali,” *Frontiers in Artificial Intelligence*, vol. 8, 2025, eight pretrained backbones; top-three MLP stacking protocol. [Online]. Available: <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1679962/full>
- [7] M. T. Ahmed, M. Rahman, S. Nur, A. Z. M. Islam, and D. Das, “Deployment of machine learning and deep learning algorithms in detecting cyberbullying in Bangla and romanized Bangla text: A comparative study,” in *2021 International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*. IEEE, 2021, pp. 1–10.
- [8] N. Romim, M. Ahmed, H. Talukder, and M. S. Islam, “Hate speech detection in the Bengali language: A dataset and its baseline evaluation,” in *Proceedings of International Joint Conference on Advances in Computational Intelligence*. Springer, 2021, pp. 457–468.
- [9] M. R. Karim, S. K. Dey, T. Islam, S. Sarker, M. H. Menon, K. Hossain, M. A. Hossain, and S. Decker, “DeepHateExplainer: Explainable hate speech detection in under-resourced Bengali language,” *arXiv preprint arXiv:2012.14353*, 2021.
- [10] A. Akhter, U. K. Acharjee, M. A. Talukder, M. M. Islam, and M. A. Uddin, “A robust hybrid machine learning model for Bengali cyber bullying detection in social media,” *Natural Language Processing Journal*, vol. 4, p. 100027, 2023.
- [11] S. Saha, M. S. Islam, M. M. Alam, M. M. Rahman, M. Z. H. Majumder, M. S. Alam, and M. K. Hossain, “Bengali cyberbullying detection in social media using machine learning algorithms,” *2023 International Conference on Electrical, Computer and Communication Engineering*, 2023.
- [12] N. Romim, M. Ahmed, M. S. Islam, A. S. Sharma, H. Talukder, and M. R. Amin, “HS-BAN: A benchmark dataset of social media comments for Bangla hate speech detection with ensemble learning,” *arXiv preprint arXiv:2112.01902*, 2021.
- [13] S. S. Nath, R. Karim, and M. H. Miraz, “Deep learning based cyberbullying detection in Bangla language,” *Annals of Emerging Technologies in Computing*, vol. 8, no. 1, pp. 50–65, 2024. [Online]. Available: <https://doi.org/10.33166/AETiC.2024.01.005>
- [14] M. I. H. Emon, K. N. Iqbal, M. H. K. Mehedi, M. J. A. Mahbub, and A. A. Rasel, “Detection of Bangla hate comments and cyberbullying in social media using NLP and transformer models,” in *Advances in Computing and Data Sciences (ICACDS 2022), CCIS 1613*. Springer, 2022, pp. 86–96.
- [15] T. T. Aurpa, R. Sadik, and M. S. Ahmed, “Abusive Bangla comments detection on Facebook using transformer-based deep learning models,” in *Social Network Analysis and Mining*, vol. 12. Springer, 2022.
- [16] K. Saifullah, M. I. Khan, S. Jamal, and I. H. Sarker, “Cyberbullying text identification based on deep learning and transformer-based language models,” *EAI Endorsed Transactions on Industrial Networks and Intelligent Systems*, vol. 11, no. 1, 2024.
- [17] M. F. Ahmed, Z. Mahmud, Z. T. Biash, A. A. N. Ryen, A. Hossain, and F. B. Ashraf, “Bangla text dataset and

- exploratory analysis for online harassment detection,” *arXiv preprint arXiv:2102.02478*, 2021. [Online]. Available: <https://arxiv.org/abs/2102.02478>
- [18] A. H. Fayaz, M. S. Uddin, R. U. Bhuiyan, Z. Sultana, M. S. Islam, B. Paul, T. Muhammad, and S. Manzoor, “BIDWESH: A Bangla regional based hate speech detection dataset,” *arXiv preprint arXiv:2507.16183*, 2025. [Online]. Available: <https://arxiv.org/abs/2507.16183>
 - [19] M. A. Hasan, F. Alam, M. F. Hossain, U. Naseem, and S. I. Ahmed, “LLM-based multi-task Bangla hate speech detection: Type, severity, and target,” *arXiv preprint arXiv:2510.01995*, 2025. [Online]. Available: <https://arxiv.org/abs/2510.01995>
 - [20] T. Haider *et al.*, “Banth: A multi-label bangla dataset for cyberbullying and hate speech,” in *Findings of the Association for Computational Linguistics: NAACL 2025*. Association for Computational Linguistics, 2025, pp. 7232–7251. [Online]. Available: <https://aclanthology.org/2025.findings-naacl.403/>
 - [21] M. M. Hasan, M. S. Arefin, M. A. Mia, and P. K. Dhar, “BanCyB: A hybrid transformer–LSTM framework for explainable multi-label cyberbullying detection in low-resource setting,” *Discover Applied Sciences*, vol. 8, no. 5, p. 575, 2026. [Online]. Available: <https://doi.org/10.1007/s42452-026-08238-4>
 - [22] M. N. Raihan, U. Tanmoy, A. B. Islam, K. North, T. Ranasinghe, A. Anastasopoulos, and M. Zampieri, “Offensive language identification in transliterated and code-mixed Bangla,” in *Proceedings of the First Workshop on Bangla Language Processing*. Association for Computational Linguistics, 2023, pp. 1–6. [Online]. Available: <https://aclanthology.org/2023.banglalp-1.1/>
 - [23] M. Fahim, F. T. Shifat, F. Haider, D. D. Barua, M. S. U. R. Sourove, M. F. Ishmam, and M. F. A. Bhuiyan, “BanglaTLit: A benchmark dataset for back-transliteration of romanized Bangla,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, 2024, pp. 14 656–14 672. [Online]. Available: <https://aclanthology.org/2024.findings-emnlp.859/>
 - [24] A. Bhattacharjee, T. Hasan, W. Ahmad, K. S. Mubasshir, M. S. Islam, A. Iqbal, M. S. Rahman, and R. Shahriyar, “BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla,” in *Findings of the Association for Computational Linguistics: NAACL 2022*. Seattle, United States: Association for Computational Linguistics, 2022, pp. 1318–1327. [Online]. Available: <https://aclanthology.org/2022.findings-naacl.98>
 - [25] S. Khanuja, D. Bansal, S. Mehtani, S. Khosla, A. Dey, B. Gopalan, D. K. Margam, P. Aggarwal, R. T. Nagipogu, S. Dave, S. Gupta, S. C. B. Gali, V. Subramanian, and P. Talukdar, “MuRIL: Multilingual representations for Indian languages,” *arXiv preprint arXiv:2103.10730*, 2021.
 - [26] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, “Unsupervised cross-lingual representation learning at scale,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 8440–8451.
 - [27] M. A. Hasan, F. Alam, M. F. Hossain, U. Naseem, and S. I. Ahmed, “Overview of BLP-2025 task 1: Bangla hate speech identification,” in *Proceedings of the Second Workshop on Bangla Language Processing*. Association for Computational Linguistics, 2025, pp. 388–401. [Online]. Available: <https://aclanthology.org/2025.banglalp-1.32/>
 - [28] T. Miyato, A. M. Dai, and I. Goodfellow, “Adversarial training methods for semi-supervised text classification,” in *International Conference on Learning Representations (ICLR)*, 2017.
 - [29] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *International Conference on Learning Representations (ICLR)*, 2015.
 - [30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.
 - [31] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, “Class-balanced loss based on effective number of samples,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 9268–9277.
 - [32] H. Inoue, “Multi-sample dropout for accelerated training and better generalization,” *arXiv preprint arXiv:1905.09788*, 2019.
 - [33] X. Liang, L. Wu, J. Li, Y. Wang, Q. Meng, T. Qin, W. Chen, M. Zhang, and T.-Y. Liu, “R-Drop: Regularized dropout for neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.
 - [34] A. Tarvainen and H. Valpola, “Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
 - [35] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, “Long-tail learning via logit adjustment,” in *International Conference on Learning Representations (ICLR)*, 2021.
 - [36] S. Sunny, R. Ahmed, A. Habib, S. Kabir, and M. A. a. Masud, “Bangla multilabel cyberbully, sexual harassment, threat and spam detection dataset,” 2024. [Online]. Available: <https://data.mendeley.com/datasets/sz5558wrd4/3>
 - [37] N. Romim, M. Ahmed, H. Talukder, M. S. Islam *et al.*, “Bd-shs: A benchmark dataset for social media hate speech detection in bangla,” in *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC 2022)*. European Language Resources Association, 2022, pp. 5153–5162. [Online]. Available: <https://aclanthology.org/2022.lrec-1.552/>
 - [38] P. Fortuna, J. Soler, and L. Wanner, “Toxic, hateful, offensive or abusive? what are we really classifying? an empirical analysis of hate speech datasets,” in *Proceedings of the Twelfth Language Resources and Evaluation Conference*. European Language Resources Association, 2020, pp. 6786–6794. [Online]. Available: <https://aclanthology.org/2020.lrec-1.838/>
 - [39] J. Cohen, “A coefficient of agreement for nominal scales,” *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

- [40] R. Artstein and M. Poesio, “Survey article: Inter-coder agreement for computational linguistics,” *Computational Linguistics*, vol. 34, no. 4, pp. 555–596, 2008. [Online]. Available: <https://aclanthology.org/J08-4004/>
- [41] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [42] K. Clark, M.-T. Luong, Q. V. Le, and C. D. Manning, “ELECTRA: Pre-training text encoders as discriminators rather than generators,” in *International Conference on Learning Representations (ICLR)*, 2020.
- [43] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations (ICLR)*, 2019.
- [44] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826.
- [45] J. A. Nelder and R. Mead, “A simplex method for function minimization,” *The Computer Journal*, vol. 7, no. 4, pp. 308–313, 1965.
- [46] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [47] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz *et al.*, “Transformers: State-of-the-art natural language processing,” *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 38–45, 2020.
- [48] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [49] B. W. Matthews, “Comparison of the predicted and observed secondary structure of T4 phage lysozyme,” *Biochimica et Biophysica Acta (BBA) - Protein Structure*, vol. 405, no. 2, pp. 442–451, 1975.
- [50] B. Efron, “Bootstrap methods: Another look at the jackknife,” *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [51] Gemini Team, Google, “Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities,” 2025. [Online]. Available: <https://arxiv.org/abs/2507.06261>